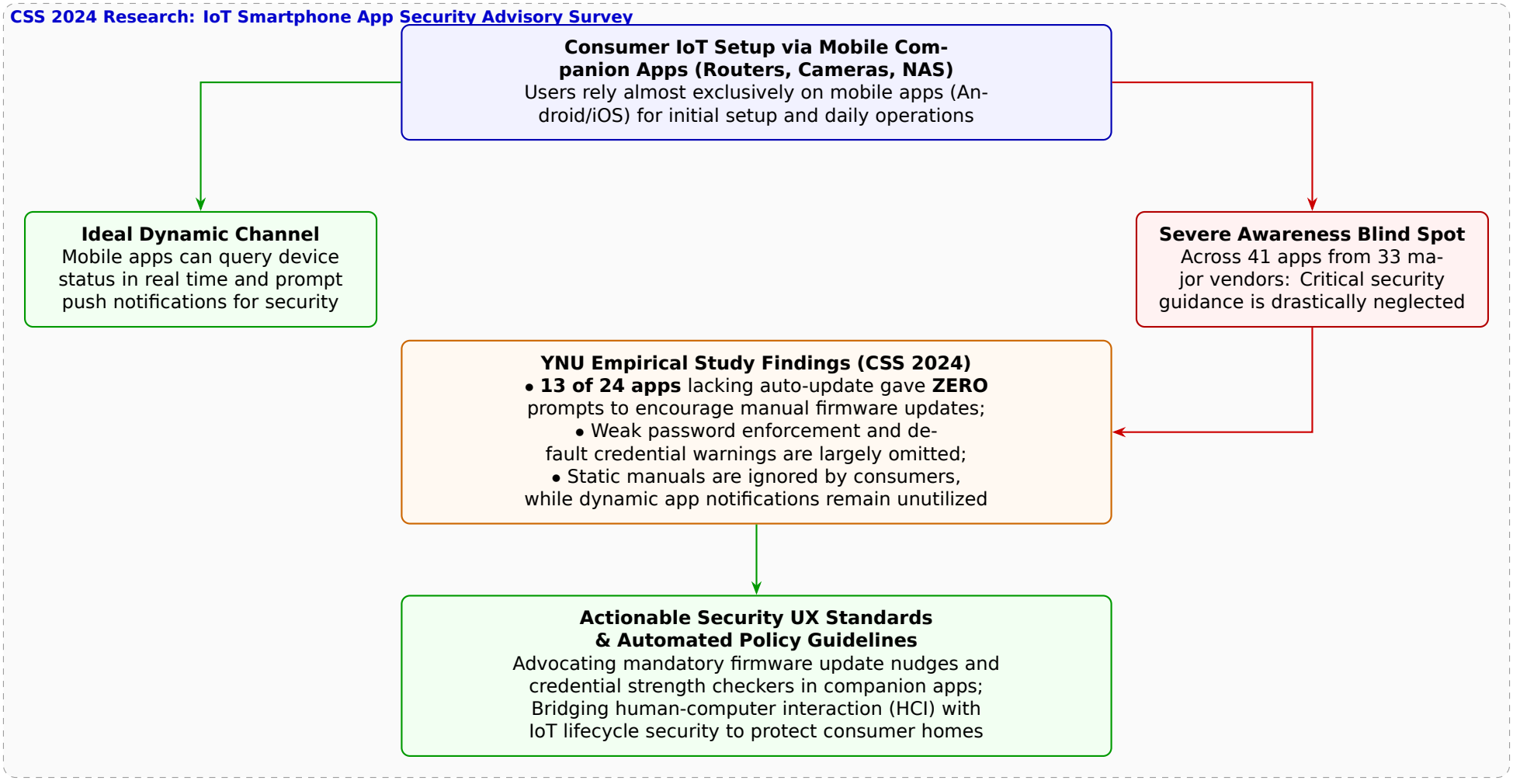
CSS 2024 Research: IoT Smartphone App Security Advisory Survey
Consumer IoT Setup via Mobile Companion Apps (Routers, Cameras, NAS)
Users rely almost exclusively on mobile apps (Android/iOS) for initial setup and daily operations
Ideal Dynamic Channel
Mobile apps can query device status in real time and prompt push notifications for security
Severe Awareness Blind Spot
Across 41 apps from 33 major vendors: Critical security guidance is drastically neglected
YNU Empirical Study Findings (CSS 2024)
• 13 of 24 apps lacking auto-update gave ZERO prompts to encourage manual firmware updates;
• Weak password enforcement and default credential warnings are largely omitted;
• Static manuals are ignored by consumers, while dynamic app notifications remain unutilized
Actionable Security UX Standards & Automated Policy Guidelines
Advocating mandatory firmware update nudges and credential strength checkers in companion apps;
Bridging human-computer interaction (HCI) with IoT lifecycle security to protect consumer homes